

Quantum Game: Schrödinger's Kittens

Daniel Magana Camargo, Ethan Gopez

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Overview. Our project, *Schrödinger's Kittens*, is a quantum computing spin on the *Exploding Kittens* card game. Inspired by Schrödinger's thought experiment, the game is centered around several kittens in a superposition of alive and dead, fighting to be the last one standing in the box.

Gameplay. Each player, or kitten, is given a qubit initially set to the $|+\rangle$ superposition and seven cards. Six of these cards are randomly drawn and can be either a gate card consisting of a standard unitary that can be applied to their own qubit and/or others', or an action card that can perform an abstract game action, such as skipping a turn, peeking at the deck, or exposing someone's qubit state, and more. The last card given to each player is a *Quantum Immortality* card that functions as a second life for the holder. Each player's turn consists of playing as many of their cards as they wish, discarding them permanently when they do so, and then drawing from the deck of shuffled cards to end their turn. This deck holds all types of cards, each with effects.

Goal. Whenever a player draws the measure card, play immediately halts and the states of every player's qubits are measured, forcing their superpositions to collapse to either $|1\rangle$ ("dead") or $|0\rangle$ ("alive"). Whenever a player measures as "alive", or whenever a player measures as "dead" and has a *Quantum Immortality* card, their qubit is reset to the $|+\rangle$ superposition, and they continue the game. If a player has been measured as "dead" and has a *Quantum Immortality* card, the card is removed from their hand. If a player measures as "dead" and does not have a *Quantum Immortality* card, they are eliminated from the game. The game continues until either only one player is left standing or none are.

Core Mechanics. The entirety of the game is implemented in a Python notebook. To play, one simply runs all cells. Most of the game is based around a universal quantum circuit filled with qubits corresponding to each player, implemented and simulated through the Qiskit library. State vector calculations and displays are also implemented with Qiskit. The random and numpy libraries are used for background calculations, while the os, time, and IPython.display libraries are used for UI purposes.

Card Types. Any number of cards may be used on a player's turn, with drawing a card indicating the end of the turn. Cards can be defined under four types.

- *Gate* cards allow a user to apply a single- or multi-qubit gate to a player's qubit of the user's choosing. They follow standard convention for the labeling of quantum gates.
 - *H gate*
 - *X/NOT gate*
 - *I gate*
 - *Y gate*
 - *Z gate*
 - *S gate*
 - *T gate*
 - *CX/CNOT gate*
 - *CZ gate*
 - *SWAP gate*
- *Measure* cards follow the logic as explained in *Goal*. Once measurement happens, the remaining players are given a new $|+\rangle$ qubit in a fresh quantum circuit in the game state. This card is immediately discarded upon draw.

- *Quantum Immortality* cards allow a player to survive a dead measurement. This card serves to be held in the user's hand until measurement and cannot be played as a move during the user's turn.
- *Action* cards allow the user to perform miscellaneous actions, many of which are analogous to the game *Exploding Kittens*.
 - *Attack*: force a player of the user's choice to discard a random card.
 - *Forced Charity*: force a player of the user's choice to insert a random card in the player's hand back into the deck at a random position.
 - *See the Future*: allows the user to view the top three cards on the deck.
 - *Shuffle*: Shuffles the deck.
 - *Favor*: force a player of the user's choice to give a random card into the player's hand to the user.
 - *Skip*: immediately ends the user's turn without drawing a card from the deck.
Allows a user to avoid unwanted cards.
 - *See Another*: allows the user to see another player's qubit of their choice.
 - *Publish Paper*: allows the user to reveal another player's qubit of their choice to the whole table.

Upon the initialization of the game state, one of every card type is inserted randomly into the deck for every player in the game. Cards are dealt as described in *Gameplay* such that no player will be dealt an extra *Quantum Immortality* or *Measure* card.

Strategy. As a general rule, the player is trying to get their qubit to the $|0\rangle$ state and other players' qubits to the $|1\rangle$ state. Furthermore, because states reset after every measurement occurrence, there is much strategy involved in whether one should use a gate to put their qubit in

a better position, or gamble and save powerful cards, like *X gates* and *Y gates*, for later measurement rounds.

Another element of strategy involves the use of gates on other players. Although a given player only knows their own qubit state, they can consider another player's moves on their own qubit to inform assumptions. For example, Player 0 may notice Player 1 uses an *H gate* card on their own qubit. This would imply Player 1 may have been in the $|+\rangle$ state and is now in the $|0\rangle$ state, or maximally the “alive” state. This could encourage Player 0 to play an *X gate* card on Player 1's qubit so that Player 1's qubit would now be maximally in $|1\rangle$, or the “dead” state. Of course, this is all based on assumption; Player 1 could've very well been in a $|-\rangle$ state and played their card to enter the $|1\rangle$ state in hopes Player 0 would play a move allowing Player 1 to enter the $|0\rangle$ state. These kinds of situations introduce elements of bluffing and signaling seen in social games of similar nature.

Furthermore, consider a situation where two players' qubits are entangled through the usage of multi-target *Gate* cards; the measurement of one qubit would have an effect on the other qubit, opening up avenues of collaboration and coercion between players entangled in the same fate. This encourages players to maintain relationships built on mutual trust whenever entering these kinds of situations.

Conclusion. *Exploding Kittens*, with its well-formulated rules and branding, has made a name for itself as a well-established party game. Through the introduction of these quantum computing elements into a well-known party game, the hope is that players of *Schrödinger's Kittens*, by playing with its mechanics, can engage in a fun and educational playspace to learn more about the fundamental principles that make up quantum computing.